Summary in Graph

Exam Summary (GO Classes GATE DA | Machine Learning | Linear and Ridge Regression | Weekly Quiz 2)



13	2	0	18m
Correct	Wrong	Skipped	of 45m

Qs. Attempted:	15 (10 + 5)
Correct Attempts:	13 (9 + 4)
Incorrect Attempts:	2 (1 + 1)

Correct Marks:	17 (9 + 8)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	17 (9 + 8)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	18 Minutes

Individual Question Statistics

View Order: Sorted Order Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (51) out of 506 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning | Easy (63%)

Assume we have $X \in \mathbb{R}^{n \times p}$ representing n data points with p features each, and $Y \in \mathbb{R}^n$ representing the outcomes.
Using linear regression with no intercept, we want to predict the outcome for a new data point $x_{\text{new}} \in \mathbb{R}^p$.
Which of the following is the correct expression for the prediction $\hat{y}_{\text{new}}$ in terms of X, Y, and x_{new} ?

A. $\hat{y}_{\text{new}} = x_{\text{new}}^T \left(X^T X\right)^{-1} X^T Y$

B. $\hat{y}_{\text{new}} = x_{\text{new}}^T XY$

C. $\hat{y}_{\text{new}} = \left(X^T X\right)^{-1} x_{\text{new}}^T Y$

D. $\hat{y}_{\text{new}} = \left(X^T X\right)^{-1} X^T Y$

Your Answer: A | Correct Answer: A | Correct | Time taken: 01min 28sec | Discuss

Q #2 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning | Medium (47%)

Suppose you want to use linear regression to fit a weight vector $w \in \mathbb{R}^d$ and an intercept term $b \in \mathbb{R}$ using training data points $x_i \in \mathbb{R}^d$.
What is the minimum number of data points n required in the training set so that the regression problem has a single unique solution?

A. $n = d$

B. $n = d + 1$

C. $n \geq d$

D. $n \geq 2d$

Your Answer: B | Correct Answer: B | Correct | Time taken: 01min 33sec | Discuss

Q #3 Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning | Hard (14%)

In ridge regression, we obtain

$$\hat{w}_{\text{ridge}} = \left(X^T X + \lambda I\right)^{-1} X^T y \quad \text{for } \lambda \geq 0$$

Which of the following statement(s) is/are TRUE?

A. $X^T X$ is always invertible.

B. $X^T X + \lambda I$ is always invertible.

C. When $\lambda = 0$, ridge regression reduces to ordinary (unregularised) linear regression.

D. As $\lambda \to \infty, \hat{w}_{\text{ridge}} \to 0$.

Your Answer: C;D | Correct Answer: C;D | Correct | Time taken: 00min 30sec | Discuss

Q #4

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning | Medium (55%)

Suppose we perform polynomial regression of degree d and degree $d+1$ (where $d \geq 1$) on the same dataset. Consider the following statements:

Statement 1: For a fixed training set, the training loss is never larger in degree $d+1$ than in degree d

Statement 2: For a fixed training and test set, the test loss is always lower in degree $d+1$ than in degree d.

Which of the following is correct?

A. Both statements are True

B. Both statements are False

C. Statement 1 is True, Statement 2 is False

D. Statement 1 is False, Statement 2 is True

Your Answer: C | Correct Answer: C | Correct | Time taken: 01min 06sec | Discuss

Q #5

Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning | Medium (39%)

In LASSO regression, consider the effect of a very large regularization parameter λ. Which of the following statement(s) is/are TRUE?

A. The model can shrink the coefficients of uninformative features to exactly 0.

B. If two informative features are highly collinear (i.e., there exists a constant α such that

$x_{ij} \approx \alpha x_{ij'}$), then LASSO may assign one of those coefficients to 0.

C. The loss function becomes the same as ridge regression when λ is very large.

D. The solution always retains all informative features with non-zero weights.

Your Answer: A;B | Correct Answer: A;B | Correct | Time taken: 00min 35sec | Discuss

Q #6

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning | Hard (-6%)

We observe a dataset consisting of input-output pairs $\{x, y\}$. Consider the following two models (assume we have unlimited data, and training/testing is done on splits of this dataset):

- Model A: $y = w^2x$
- Model B: $y = wx$

Which of the following is correct?

A. There exist datasets for which Model A would perform better than Model B.

B. There exist datasets for which Model B would perform better than Model A.

C. Both A and B are correct.

D. They would perform equally well on all datasets.

Your Answer: B | Correct Answer: B | Correct | Time taken: 01min 03sec | Discuss

Q #7

Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning | Medium (39%)

In a simple linear regression model with one independent variable, if we increase the input variable by 1 unit, how much will the output variable change?

A. By its slope

B. By its intercept

C. By 1

D. No change

Your Answer: A | Correct Answer: A | Correct | Time taken: 00min 33sec | Discuss

Q #8 Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning | Medium (48%)

Select all statements that are true.

A. If there are two identical features in the data, the L^2-regularization will force the coefficient of one redundant feature to be 0.

B. We cannot use linear regression to find the coefficients for θ in $y = \theta_1 x^3 + \theta_2 x^2 + \theta_3 x + \theta_4$ since the relationship between y and x is non-linear.

C. Introducing more features increases the model complexity and may cause overfitting.

D. None of the above statements are true.

Your Answer: B;C | Correct Answer: C | Incorrect | Time taken: 00min 49sec | Discuss

Q #9 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning | Medium (39%)

In simple linear regression with an intercept, $\hat{y} = w_0 + w_1 x$.
Let $\bar{x}$ and $\bar{y}$ denote the sample means:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

Which statement is correct?

A. The point $(\bar{x}, \bar{y})$ always lies on the fitted line.

B. The point ($\bar{x}, \bar{y}$) lies on the fitted line only if $\bar{x} = 0$.

C. The point $(\bar{x}, \bar{y})$ never lies on the fitted line.

D. Whether $(\bar{x}, \bar{y})$ lies on the line depends only on the slope estimate.

Your Answer: A | Correct Answer: A | Correct | Time taken: 00min 34sec | Discuss

Q #10 Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning | Hard (27%)

Which of the following are guaranteed to be true regarding un-regularized ordinary least squares linear regression with an intercept term?

A. The mean of the residuals is 0.

B. The mean of the fitted values $\hat{y}$ equals the mean of the observed values y.

C. The weight corresponding to the intercept term is positive.

D. There is a unique solution to the normal equations.

Your Answer: A;B | Correct Answer: A;B | Correct | Time taken: 01min 04sec | Discuss

Q #11

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning | Easy (61%)

Consider the ridge regression optimization problem:

$$\min_{\theta \in \mathbb{R}^d} \frac{1}{2}\|X\theta - y\|^2 + \frac{\lambda}{2}\|\theta\|^2,$$

where $X \in \mathbb{R}^{n \times d}, y \in \mathbb{R}^n, \lambda > 0$.
Which of the following statements is TRUE?

A. Ridge regression may not have a unique solution if X has a null space.

B. Ridge regression always has a unique solution if $\lambda > 0$, even when X has a null space.

C. Ridge regression solution does not depend on λ.

D. Ridge regression is equivalent to ordinary least squares in all cases.

Your Answer: B | Correct Answer: B | Correct | Time taken: 01min 13sec | Discuss

Q #12

Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning | Hard (20%)

Consider the Least Squares estimator

$$\hat{\beta} = \left(X^T X\right)^{-1} X^T y$$

with residual vector $e = y - \hat{y}$, where $\hat{y} = X\hat{\beta}$.

Which of the following statement(s) is/are TRUE?

A. There exists a set of weights β such that

$$\sum_{i=1}^{n} \left(y_i - (X\beta)_i\right)^2 < \sum_{i=1}^{n} e_i^2.$$

B. The residual vector e is always orthogonal to the fitted values $\hat{y}$ (i.e., $e^T \hat{y} = 0$).

C. For any set of weights β, the residual vector e is orthogonal to $X(\hat{\beta} - \beta)$.

D. None of the above.

Your Answer: B;C | Correct Answer: B;C | Correct | Time taken: 02min 02sec | Discuss

Q #13

Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning | Hard (17%)

We have data $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ and fit the simple linear model

$$y = w_0 + w_1 x.$$

The design matrix is

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \in \mathbb{R}^{n \times 2}, \quad w = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad X^T X = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix}.$$

Which of the following statement(s) is/are correct?

A. The sample size is $n = 4$.

B. $X^T y = \begin{bmatrix} 18 \\ 29 \end{bmatrix}$.

C. The sample size is $n = 2$.

D. $X^T y$ cannot be determined from the given information.

Your Answer: A | Correct Answer: A;B | Incorrect | Time taken: 00min 26sec | Discuss

Q #14

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning | Medium (32%)

Let $X \in \mathbb{R}^{n \times (p+1)}$ (first column all ones for the intercept; remaining p columns are features) and $y \in \mathbb{R}^n$. Consider the two ridge estimators:

$$\hat{\theta}_A = \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} \left(y_i - x_i^\top \theta\right)^2 + \lambda \sum_{k=0}^{p} \theta_k^2 \quad \text{(intercept penalized)}$$

$$\hat{\theta}_B = \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} \left(y_i - x_i^\top \theta\right)^2 + \lambda \sum_{k=1}^{p} \theta_k^2 \quad \text{(intercept not penalized)}$$

As $\lambda \to \infty$, the predictions from these models approach which limits?

A. $\hat{\theta}_A \to 0, \quad \hat{\theta}_B \to 0$

B. $\hat{\theta}_A \to 0, \quad \hat{\theta}_B \to \bar{y}$ (the sample mean of the true y values)

C. $\hat{\theta}_A \to \bar{y}, \quad \hat{\theta}_B \to 0$

D. $\hat{\theta}_A \to \bar{y}, \quad \hat{\theta}_B \to \bar{y}$

Your Answer: B | Correct Answer: B | Correct | Time taken: 04min 36sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning | Medium (32%)

Let $X \in \mathbb{R}^{n \times d}$ and $y \in \mathbb{R}^n$. In ordinary least squares linear regression, we define

$$\hat{w} = \left(X^\top X\right)^{-1} X^\top y, \quad \hat{y} = X\hat{w}, \quad r = y - \hat{y}, \quad H = X\left(X^\top X\right)^{-1} X^\top$$

Here, $\hat{y}$ is the vector of predictions, r is the residual vector, and H is known as the projection matrix .

Which of the following statements is/are true?

A. $Hy = \hat{y}$

B. $H\hat{y} = \hat{y}$

C. $Hy = r$

D. $H\hat{y} = 0$

Your Answer: A;B | Correct Answer: A;B | Correct | Time taken: 00min 45sec | Discuss

You're doing Great!

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